



# Machine Learning WS 2020 Final Exam Solution

Machine learning (Technische Universität München)



Scanne, um auf Studocu zu öffnen

**Ecorrection**

Place student sticker here

**Note:**

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

## Machine Learning

**Graded Exercise:** IN2064 / Endterm  
**Examiner:** Prof. Dr. Stephan Günnemann

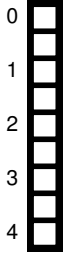
**Date:** Tuesday 16<sup>th</sup> February, 2021  
**Time:** 11:00 – 13:00

### Working instructions

- This graded exercise consists of **28 pages** with a total of **23 problems**. Please make sure now that you received a complete copy of the answer sheet.
- The total amount of achievable credits in this graded exercise is 107 credits.
- Allowed resources:
  - all materials that you will use on your own (lecture slides, calculator etc.)
  - **not allowed are any forms of collaboration between examinees and plagiarism**
- You have to sign the code of conduct. (Typing your name is fine)
- You have to either print this document and scan your solutions or paste scans/pictures of your handwritten solutions into the solution boxes in this PDF. **Editing the PDF digitally is prohibited except for signing the code of conduct and answering multiple choice questions.**
- Make sure that the **QR codes are visible** on every uploaded page. Otherwise, we cannot grade your submission.
- **You must solve the specified version of the problem.** Different problems may have different version: e.g. Problem 1 (Version A), Problem 5 (Version C), etc. If you solve the wrong version you get **zero** points.
- Only write on the provided sheets, **submitting your own additional sheets is not possible**.
- Last three pages can be used as scratch paper.
- All sheets (including scratch paper) have to be submitted to the upload queue. Missing pages will be considered empty.
- **Only use a black or blue color (no red or green)! Pencils are allowed.**
- Write your answers only in the provided solution boxes or the scratch paper.
- **For problems that say "Justify your answer" you only get points if you provide a valid explanation.**
- **For problems that say "Prove" you only get points if you provide a valid mathematical proof.**
- If a problem does not say "Justify your answer" or "Prove" it's sufficient to only provide the correct answer.
- Instructor announcements and clarifications will be posted **on Piazza** with email notifications.
- Exercise duration - 120 minutes.

Left room from \_\_\_\_\_ to \_\_\_\_\_ / Early submission at \_\_\_\_\_

## Problem 1 (Version A) (4 credits)



We have to find the most likely value  $s^*$  of  $s$  after incorporating the observations of  $t$ , i.e. the maximum a posteriori estimate.

$$\begin{aligned}
 s^* &= \arg \max_s p(\mathcal{D} | s) p(s) \\
 &= \arg \max_s \log p(\mathcal{D} | s) + \log p(s) \\
 &= \arg \max_s \sum_{i=1}^N \log s^2 \exp(-s^2 t_i) + \log \exp(-s^2) \\
 &= \arg \max_s N \log s^2 - s^2 \sum_{i=1}^N t_i - s^2 \\
 &= \arg \max_s N \log s^2 - s^2(T + 1) \text{ where } T = \sum_{i=1}^N t_i
 \end{aligned}$$

This expression is symmetric in the sign of  $s$ , so we can restrict ourselves to the case of  $s \geq 0$ . On this restricted domain, the expression is also concave in  $s$ , so we can find the maximum by differentiation.

$$\frac{\partial}{\partial s} N \log s^2 - s^2(T + 1) = \frac{2N}{s} - 2(T + 1)s = 0 \Leftrightarrow s = \pm \sqrt{\frac{N}{T + 1}}$$

Summing the observations, we get  $T = 19$  and so the positive most likely severity of the disease is  $s^* = \sqrt{\frac{5}{19+1}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$ .

*Note:* The problem description had a small mistake and depending on if the students worked with  $\exp(-s^2)$  or  $\exp\left(-\frac{s^2}{2}\right)$ , the students might also have arrived at

$$s^* = \arg \max_s N \log s^2 - s^2\left(T + \frac{1}{2}\right).$$

Then their end result would be  $s^* = \sqrt{\frac{5}{19+\frac{1}{2}}} = \sqrt{\frac{10}{39}} \approx 0.506$ .

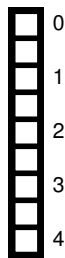
✓ for recognizing this as a maximum a-posteriori problem with prior and likelihood, i.e. by writing  $s^* = \arg \max_s p(\mathcal{D} | s) p(s)$  or similar

Up to ✓✓✓ for the remainder of the calculation depending on how right/wrong it is

If the most likely severity  $s^*$  is wrong, subtract at least  $\frac{1}{2}$  ✓

If likelihood is not taken as a product over all  $t_i$  (i.e. unclear/missing dependence on the number of and sum of data points), subtract at least another ✓

## Problem 1 (Version B) (4 credits)



We have to find the most likely value  $s^*$  of  $s$  after incorporating the observations of  $t$ , i.e. the maximum a posteriori estimate.

$$\begin{aligned} s^* &= \arg \max_s p(\mathcal{D} | s) p(s) \\ &= \arg \max_s \log p(\mathcal{D} | s) + \log p(s) \\ &= \arg \max_s \sum_{i=1}^N \log s^2 \exp(-s^2 t_i) + \log \exp(-s^2) \\ &= \arg \max_s N \log s^2 - s^2 \sum_{i=1}^N t_i - s^2 \\ &= \arg \max_s N \log s^2 - s^2 (T + 1) \text{ where } T = \sum_{i=1}^N t_i \end{aligned}$$

This expression is symmetric in the sign of  $s$ , so we can restrict ourselves to the case of  $s \geq 0$ . On this restricted domain, the expression is also concave in  $s$ , so we can find the maximum by differentiation.

$$\frac{\partial}{\partial s} N \log s^2 - s^2 (T + 1) = \frac{2N}{s} - 2(T + 1)s = 0 \Leftrightarrow s = \pm \sqrt{\frac{N}{T + 1}}$$

Summing the observations, we get  $T = 26$  and so the positive most likely severity of the disease is  $s^* = \sqrt{\frac{3}{26+1}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$ .

*Note:* The problem description had a small mistake and depending on if the students worked with  $\exp(-s^2)$  or  $\exp\left(-\frac{s^2}{2}\right)$ , the students might also have arrived at

$$s^* = \arg \max_s N \log s^2 - s^2 \left(T + \frac{1}{2}\right).$$

Then their end result would be  $s^* = \sqrt{\frac{3}{26+\frac{1}{2}}} = \sqrt{\frac{6}{53}} \approx 0.336$ .

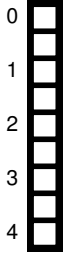
✓ for recognizing this as a maximum a-posteriori problem with prior and likelihood, i.e. by writing  $s^* = \arg \max_s p(\mathcal{D} | s) p(s)$  or similar

Up to ✓✓✓ for the remainder of the calculation depending on how right/wrong it is

If the most likely severity  $s^*$  is wrong, subtract at least  $\frac{1}{2}$  ✓

If likelihood is not taken as a product over all  $t_i$  (i.e. unclear/missing dependence on the number of and sum of data points), subtract at least another ✓

## Problem 1 (Version C) (4 credits)



We have to find the most likely value  $s^*$  of  $s$  after incorporating the observations of  $t$ , i.e. the maximum a posteriori estimate.

$$\begin{aligned}
 s^* &= \arg \max_s p(\mathcal{D} | s) p(s) \\
 &= \arg \max_s \log p(\mathcal{D} | s) + \log p(s) \\
 &= \arg \max_s \sum_{i=1}^N \log s^2 \exp(-s^2 t_i) + \log \exp(-s^2) \\
 &= \arg \max_s N \log s^2 - s^2 \sum_{i=1}^N t_i - s^2 \\
 &= \arg \max_s N \log s^2 - s^2(T + 1) \text{ where } T = \sum_{i=1}^N t_i
 \end{aligned}$$

This expression is symmetric in the sign of  $s$ , so we can restrict ourselves to the case of  $s \geq 0$ . On this restricted domain, the expression is also concave in  $s$ , so we can find the maximum by differentiation.

$$\frac{\partial}{\partial s} N \log s^2 - s^2(T + 1) = \frac{2N}{s} - 2(T + 1)s = 0 \Leftrightarrow s = \pm \sqrt{\frac{N}{T + 1}}$$

Summing the observations, we get  $T = 35$  and so the positive most likely severity of the disease is  $s^* = \sqrt{\frac{4}{35+1}} = \sqrt{\frac{1}{9}} = \frac{1}{3}$ .

*Note:* The problem description had a small mistake and depending on if the students worked with  $\exp(-s^2)$  or  $\exp\left(-\frac{s^2}{2}\right)$ , the students might also have arrived at

$$s^* = \arg \max_s N \log s^2 - s^2\left(T + \frac{1}{2}\right).$$

Then their end result would be  $s^* = \sqrt{\frac{4}{35+\frac{1}{2}}} = \sqrt{\frac{8}{71}} \approx 0.336$ .

✓ for recognizing this as a maximum a-posteriori problem with prior and likelihood, i.e. by writing  $s^* = \arg \max_s p(\mathcal{D} | s) p(s)$  or similar

Up to ✓ ✓ ✓ for the remainder of the calculation depending on how right/wrong it is

If the most likely severity  $s^*$  is wrong, subtract at least  $\frac{1}{2}$  ✓

If likelihood is not taken as a product over all  $t_i$  (i.e. unclear/missing dependence on the number of and sum of data points), subtract at least another ✓

## Problem 2 (Version A) (4 credits)

a)

The value of  $k = 5$  minimizes the LOOCV error. ✓ The error is  $4/13$ . ✓

- 1 ✓ for the correct value of  $k$ .
- 1 ✓ for the correct error. Both 4 and  $4/13$  are valid.

0  
1  
2

b)

Yes. One counter example is that  $(1, 3)$  was previously labeled with  $-$  but is now labeled with  $+$  since the new point  $(1, 2)$  is closest. ✓

- 1 ✓ for a valid reason why the decision boundary has changed. A single counter example for which the decision boundary was changed is valid.
- 0.5 ✓ for "Yes" with partially correct justification. Zero points for no justification or completely wrong justification.
- A common mistake was to consider the prediction changes *only* for the training data. However, since the question is about the decision boundary you have to think about the *entire* input space.

0  
1

c)

We need to move it to the closest data point with the same label to keep the decision boundary the same. In this case this is  $(5, 1)$  ✓. The distance is  $\sqrt{1^2 + 4^2} = \sqrt{17}$ . ✓

- 0.5 ✓ for a correct new position – the closest training point of the same class.
- 0.5 ✓ for the correct distance.

0  
1

## Problem 2 (Version B) (4 credits)

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |
| 2 | <input type="checkbox"/> |

a)

The value of  $k = 5$  minimizes the LOOCV error. ✓ The error is  $4/13$ . ✓

- 1 ✓ for the correct value of  $k$ .
- 1 ✓ for the correct error. Both 4 and  $4/13$  are valid.

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |

b)

Yes. One counter example is that  $(2, 2)$  was previously labeled with  $\ominus$  but is now labeled with  $\oplus$  since the new point  $(1, 2)$  is closest. ✓

- 1 ✓ for a valid reason why the decision boundary has changed. A single counter example for which the decision boundary was changed is valid.
- 0.5 ✓ for "Yes" with partially correct justification. Zero points for no justification or completely wrong justification.
- A common mistake was to consider the prediction changes *only* for the training data. However, since the question is about the decision boundary you have to think about the *entire* input space.

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |

c)

We need to move it to the closest data point with the same label to keep the decision boundary the same. In this case this is  $(2, 6)$  ✓. The distance is  $\sqrt{1^2 + 4^2} = \sqrt{17}$ . ✓

- 0.5 ✓ for a correct new position – the closest training point of the same class.
- 0.5 ✓ for the correct distance.

## Problem 2 (Version C) (4 credits)

a)

The value of  $k = 5$  minimizes the LOOCV error. ✓ The error is  $4/13$ . ✓

- 1 ✓ for the correct value of  $k$ .
- 1 ✓ for the correct error. Both 4 and  $4/13$  are valid.

0  
1  
2

b)

Yes. One counter example is that  $(2, 2)$  was previously labeled with  $+$  but is now labeled with  $-$  since the new point  $(1, 2)$  is closest. ✓

- 1 ✓ for a valid reason why the decision boundary has changed. A single counter example for which the decision boundary was changed is valid.
- 0.5 ✓ for "Yes" with partially correct justification. Zero points for no justification or completely wrong justification.
- A common mistake was to consider the prediction changes *only* for the training data. However, since the question is about the decision boundary you have to think about the *entire* input space.

0  
1

c)

We need to move it to the closest data point with the same label to keep the decision boundary the same. In this case this is  $(2, 6)$  ✓. The distance is  $\sqrt{1^2 + 4^2} = \sqrt{17}$ . ✓

- 0.5 ✓ for a correct new position – the closest training point of the same class.
- 0.5 ✓ for the correct distance.

0  
1

## Problem 2 (Version D) (4 credits)

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |
| 2 | <input type="checkbox"/> |

a)

The value of  $k = 5$  minimizes the LOOCV error. ✓ The error is  $4/13$ . ✓

- 1 ✓ for the correct value of  $k$ .
- 1 ✓ for the correct error. Both 4 and  $4/13$  are valid.

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |

b)

Yes. One counter example is that  $(1, 3)$  was previously labeled with  $+$  but is now labeled with  $-$  since the new point  $(1, 2)$  is closest. ✓

- 1 ✓ for a valid reason why the decision boundary has changed. A single counter example for which the decision boundary was changed is valid.
- 0.5 ✓ for "Yes" with partially correct justification. Zero points for no justification or completely wrong justification.
- A common mistake was to consider the prediction changes *only* for the training data. However, since the question is about the decision boundary you have to think about the *entire* input space.

|   |                          |
|---|--------------------------|
| 0 | <input type="checkbox"/> |
| 1 | <input type="checkbox"/> |

c)

We need to move it to the closest data point with the same label to keep the decision boundary the same. In this case this is  $(5, 1)$  ✓. The distance is  $\sqrt{1^2 + 4^2} = \sqrt{17}$ . ✓

- 0.5 ✓ for a correct new position – the closest training point of the same class.
- 0.5 ✓ for the correct distance.

### Problem 3 (Version A) (6 credits)

a)

Because of convexity, we can find the optimal  $w_{D+1}$  by finding the zero of the derivative.

$$\begin{aligned}\frac{\partial}{\partial w_{D+1}} J(\mathbf{w}) &= \sum_{i=1}^N (\mathbf{w}^T \tilde{\mathbf{x}}^{(i)} - y^{(i)}) + \lambda w_{D+1} \\ &= \mathbf{w}_{1:D}^T \sum_{i=1}^N \mathbf{x}^{(i)} + w_{D+1} \sum_{i=1}^N 1 - \sum_{i=1}^N y^{(i)} + \lambda w_{D+1}\end{aligned}$$

$\sum_{i=1}^N \mathbf{x}^{(i)}$  is zero because we have assumed that the  $\mathbf{x}^i$  are centered.

$$= Nw_{D+1} - \sum_{i=1}^N y^{(i)} + \lambda w_{D+1} = (N + \lambda)w_{D+1} - \sum_{i=1}^N y^{(i)}$$

Solving for  $w_{D+1}$  we get

$$w_{D+1} = \frac{1}{N + \lambda} \sum_{i=1}^N y^{(i)}.$$

At most ✓ for any solution that is just stating the normal equation  $\tilde{\mathbf{w}} = (\tilde{X}^T \tilde{X} + \lambda \mathbf{I})^{-1} \tilde{X}^T \mathbf{y}$  or similar  
Otherwise up to ✓ ✓ ✓ depending on correctness

b)

We propose a biased centering of the regression targets, i.e.

$$\tilde{\mathbf{x}}^{(i)} = \mathbf{x}^{(i)} \quad \text{and} \quad \tilde{y}^{(i)} = y^{(i)} - \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)}.$$

The ridge regression loss evaluated on  $\tilde{\mathcal{D}}$  is

$$\mathcal{L}(\tilde{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N (\tilde{\mathbf{w}}_{1:D}^T \mathbf{x}^{(i)} + \tilde{w}_{D+1} - y^{(i)})^2 + \frac{\lambda}{2} \|\tilde{\mathbf{w}}\|_2^2 + \frac{\lambda}{2} \tilde{w}_{D+1}^2.$$

The gradient and therefore the optimal value of  $\tilde{\mathbf{w}}_{D+1}$  is independent of  $\tilde{\mathbf{w}}_{1:D}$ , so for the optimal values of  $\tilde{\mathbf{w}}_{1:D}$  it is equivalent to minimize  $\mathcal{L}$  with  $\tilde{\mathbf{w}}_{D+1}^*$  plugged in.

$$\mathcal{L}(\tilde{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N \left( \tilde{\mathbf{w}}_{1:D}^T \mathbf{x}^{(i)} + \left( \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)} \right) - y^{(i)} \right)^2 + \frac{\lambda}{2} \|\tilde{\mathbf{w}}_{1:D}\|_2^2 + \frac{\lambda}{2} \left( \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)} \right)^2.$$

The last part has zero gradient with respect to  $\tilde{\mathbf{w}}_{1:D}$ , so it does not influence the optimal  $\tilde{\mathbf{w}}_{1:D}^*$  and we can drop it since  $\tilde{\mathbf{w}}_{D+1}$  has been eliminated. If we then absorb the  $\frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)}$  term in the least squares regression sum into  $y^{(i)}$ , we get the ridge regression loss evaluated on  $\tilde{\mathcal{D}}$

$$\mathcal{L}(\hat{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N (\hat{\mathbf{w}}^T \mathbf{x}^{(i)} - \hat{y}^{(i)})^2 + \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2.$$

showing that ridge regression on  $\hat{\mathcal{D}}$  is equivalent to ridge regression on  $\tilde{\mathcal{D}}$ .  $\frac{1}{2}$  ✓ for proposing to center the regression targets in any way, ✓ for the correct preprocessing step/rule

Up to ✓ ✓ for the justification for why ridge regression finds the same first  $D$  weights

0 ✓ for any solution that violates  $\hat{\mathbf{w}} \in \mathbb{R}^{D(D+1)}$

### Problem 3 (Version B) (6 credits)

a)

Because of convexity, we can find the optimal  $w_{D+1}$  by finding the zero of the derivative.

$$\begin{aligned}\frac{\partial}{\partial w_{D+1}} J(\mathbf{w}) &= \sum_{i=1}^N (\mathbf{w}^T \tilde{\mathbf{x}}^{(i)} - y^{(i)}) + \lambda w_{D+1} \\ &= \mathbf{w}_{1:D}^T \sum_{i=1}^N \mathbf{x}^{(i)} + w_{D+1} \sum_{i=1}^N 1 - \sum_{i=1}^N y^{(i)} + \lambda w_{D+1}\end{aligned}$$

$\sum_{i=1}^N \mathbf{x}^{(i)}$  is zero because we have assumed that the  $\mathbf{x}^i$  are centered.

$$= Nw_{D+1} - \sum_{i=1}^N y^{(i)} + \lambda w_{D+1} = (N + \lambda)w_{D+1} - \sum_{i=1}^N y^{(i)}$$

Solving for  $w_{D+1}$  we get

$$w_{D+1} = \frac{1}{N + \lambda} \sum_{i=1}^N y^{(i)}.$$

At most ✓ for any solution that is just stating the normal equation  $\tilde{\mathbf{w}} = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{X}}^T \mathbf{y}$  or similar  
Otherwise up to ✓ ✓ ✓ depending on correctness

b)

We propose a biased centering of the regression targets, i.e.

$$\hat{\mathbf{x}}^{(i)} = \mathbf{x}^{(i)} \quad \text{and} \quad \hat{y}^{(i)} = y^{(i)} - \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)}.$$

The ridge regression loss evaluated on  $\tilde{\mathcal{D}}$  is

$$\mathcal{L}(\tilde{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N (\tilde{\mathbf{w}}_{1:D}^T \mathbf{x}^{(i)} + \tilde{w}_{D+1} - y^{(i)})^2 + \frac{\lambda}{2} \|\tilde{\mathbf{w}}\|_2^2 + \frac{\lambda}{2} \tilde{w}_{D+1}^2.$$

The gradient and therefore the optimal value of  $\tilde{w}_{D+1}$  is independent of  $\tilde{\mathbf{w}}_{1:D}$ , so for the optimal values of  $\tilde{\mathbf{w}}_{1:D}$  it is equivalent to minimize  $\mathcal{L}$  with  $\tilde{w}_{D+1}^*$  plugged in.

$$\mathcal{L}(\tilde{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N \left( \tilde{\mathbf{w}}_{1:D}^T \mathbf{x}^{(i)} + \left( \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)} \right) - y^{(i)} \right)^2 + \frac{\lambda}{2} \|\tilde{\mathbf{w}}_{1:D}\|_2^2 + \frac{\lambda}{2} \left( \frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)} \right)^2.$$

The last part has zero gradient with respect to  $\tilde{\mathbf{w}}_{1:D}$ , so it does not influence the optimal  $\tilde{\mathbf{w}}_{1:D}^*$  and we can drop it since  $\tilde{w}_{D+1}$  has been eliminated. If we then absorb the  $\frac{1}{N + \lambda} \sum_{j=1}^N y^{(j)}$  term in the least squares regression sum into  $y^{(i)}$ , we get the ridge regression loss evaluated on  $\hat{\mathcal{D}}$

$$\mathcal{L}(\hat{\mathbf{w}}) = \frac{1}{2} \sum_{i=1}^N (\hat{\mathbf{w}}^T \mathbf{x}^{(i)} - \hat{y}^{(i)})^2 + \frac{\lambda}{2} \|\hat{\mathbf{w}}\|_2^2.$$

showing that ridge regression on  $\hat{\mathcal{D}}$  is equivalent to ridge regression on  $\tilde{\mathcal{D}}$ .  $\frac{1}{2}$  ✓ for proposing to center the regression targets in any way, ✓ for the correct preprocessing step/rule

Up to ✓ ✓ for the justification for why ridge regression finds the same first  $D$  weights

0 ✓ for any solution that violates  $\hat{\mathbf{w}} \in \mathbb{R}^{D \neq D+1}$

## Problem 4 (Version A) (6 credits)

a)

In a naive Bayes classifier, the features are independent, so we can choose a different probability distribution for each of them. We choose to model the continuous feature as a normal distribution,  $x_1 | y = c \sim \mathcal{N}(\mu_c, 1)$ . The discrete feature can be one of two values which we model with a Bernoulli distribution  $x_3 | y = c \sim \text{Bernoulli}(\alpha_c)$  where yes is 1, no is 0 and  $\alpha_c$  gives the success probability. The distribution of the classes  $y$  is a categorical distribution with parameter  $\pi$ ,  $y \sim \text{Categorical}(\pi)$ . The maximum likelihood estimates of the parameters are

$$\pi = \begin{pmatrix} \frac{2}{7} & \frac{3}{7} & \frac{2}{7} \end{pmatrix}^T$$

$$\mu_1 = 1 \quad \mu_2 = 0 \quad \mu_3 = 5$$

$$\alpha_1 = \frac{1}{2} \quad \alpha_2 = \frac{1}{3} \quad \alpha_3 = 1$$

- $\frac{1}{2}$  ✓ for modelling the class prior
- ✓ for modelling the two data attributes independently
- $\frac{1}{2}$  ✓ for computing  $\pi$  correctly
- $\frac{1}{2}$  ✓ for computing  $\mu_1, \dots, \mu_3$  correctly
- $\frac{1}{2}$  ✓ for computing  $\alpha_1, \dots, \alpha_3$  correctly

b)

The unnormalized posterior is  $p(y^{(b)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_1^{(b)} | y^{(b)}) p(\mathbf{x}_2^{(b)} | y^{(b)}) p(y^{(b)})$ , so we evaluate that for all three choices of  $y^{(b)}$  and get

$$p(y^{(b)} | \mathbf{x}^{(b)}) \propto \begin{pmatrix} e^{0 \cdot \frac{1}{2} \cdot \frac{2}{7}} & e^{-\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{7}} & e^{-8 \cdot 1 \cdot \frac{2}{7}} \end{pmatrix}^T = \begin{pmatrix} \frac{1}{7} & \frac{1}{7\sqrt{e}} & \frac{2}{7e^8} \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

c)

We do not know anything about this data point, so the posterior distribution is just the prior distribution.

$$p(y^{(c)} | \mathbf{x}^{(c)}) = p(y) = \begin{pmatrix} \frac{2}{7} & \frac{3}{7} & \frac{2}{7} \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

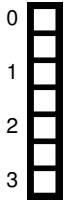
d)

Since we only know the feature  $x_2^{(d)}$ , we only condition on that and get  $p(y^{(d)} | \mathbf{x}^{(d)}) \propto p(\mathbf{x}_2^{(b)} | y^{(d)}) p(y^{(d)})$ .

$$p(y^{(d)} | \mathbf{x}^{(d)}) = \begin{pmatrix} \frac{1}{2} \cdot \frac{2}{7} & \frac{2}{3} \cdot \frac{3}{7} & 0 \cdot \frac{2}{7} \end{pmatrix}^T = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} & 0 \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

## Problem 4 (Version B) (6 credits)



a)

In a naive Bayes classifier, the features are independent, so we can choose a different probability distribution for each of them. We choose to model the continuous feature as a normal distribution,  $x_1 | y = c \sim \mathcal{N}(\mu_c, 1)$ . The discrete feature can be one of two values which we model with a Bernoulli distribution  $x_3 | y = c \sim \text{Bernoulli}(\alpha_c)$  where yes is 1, no is 0 and  $\alpha_c$  gives the success probability. The distribution of the classes  $y$  is a categorical distribution with parameter  $\pi$ ,  $y \sim \text{Categorical}(\pi)$ . The maximum likelihood estimates of the parameters are

$$\pi = \left( \frac{2}{7} \quad \frac{3}{7} \quad \frac{2}{7} \right)^T$$

$$\mu_1 = 1 \quad \mu_2 = 0 \quad \mu_3 = 5$$

$$\alpha_1 = \frac{1}{2} \quad \alpha_2 = \frac{1}{3} \quad \alpha_3 = 1$$

- $\frac{1}{2}$  ✓ for modelling the class prior
- ✓ for modelling the two data attributes independently
- $\frac{1}{2}$  ✓ for computing  $\pi$  correctly
- $\frac{1}{2}$  ✓ for computing  $\mu_1, \dots, \mu_3$  correctly
- $\frac{1}{2}$  ✓ for computing  $\alpha_1, \dots, \alpha_3$  correctly



b)

The unnormalized posterior is  $p(y^{(b)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_1^{(b)} | y^{(b)}) p(\mathbf{x}_2^{(b)} | y^{(b)}) p(y^{(b)})$ , so we evaluate that for all three choices of  $y^{(b)}$  and get

$$p(y^{(b)} | \mathbf{x}^{(b)}) \propto \left( e^{-\frac{1}{2} \frac{1}{2} \frac{2}{7}} \quad e^{-\frac{1}{2} \frac{1}{3} \frac{3}{7}} \quad e^{-\frac{9}{2} \frac{2}{7}} \right)^T = \left( \frac{1}{7\sqrt{e}} \quad \frac{1}{7e^2} \quad \frac{2}{7e^{\frac{9}{2}}} \right)^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).



c)

We do not know anything about this data point, so the posterior distribution is just the prior distribution.

$$p(y^{(c)} | \mathbf{x}^{(c)}) = p(y) = \left( \frac{2}{7} \quad \frac{3}{7} \quad \frac{2}{7} \right)^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).



d)

Since we only know the feature  $x_2^{(d)}$ , we only condition on that and get  $p(y^{(d)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_2^{(b)} | y^{(d)}) p(y^{(d)})$ .

$$p(y^{(d)} | \mathbf{x}^{(d)}) = \left( \frac{1}{2} \frac{2}{7} \quad \frac{2}{3} \frac{3}{7} \quad 0 \frac{2}{7} \right)^T = \left( \frac{1}{7} \quad \frac{2}{7} \quad 0 \right)^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

## Problem 4 (Version C) (6 credits)

a)

In a naive Bayes classifier, the features are independent, so we can choose a different probability distribution for each of them. We choose to model the continuous feature as a normal distribution,  $x_1 | y = c \sim \mathcal{N}(\mu_c, 1)$ . The discrete feature can be one of two values which we model with a Bernoulli distribution  $x_3 | y = c \sim \text{Bernoulli}(\alpha_c)$  where yes is 1, no is 0 and  $\alpha_c$  gives the success probability. The distribution of the classes  $y$  is a categorical distribution with parameter  $\pi$ ,  $y \sim \text{Categorical}(\pi)$ . The maximum likelihood estimates of the parameters are

$$\pi = \begin{pmatrix} \frac{2}{7} & \frac{2}{7} & \frac{3}{7} \end{pmatrix}^T$$

$$\mu_1 = -2 \quad \mu_2 = 2 \quad \mu_3 = 4$$

$$\alpha_1 = \frac{1}{2} \quad \alpha_2 = 0 \quad \alpha_3 = \frac{2}{3}$$

- $\frac{1}{2}$  ✓ for modelling the class prior
- ✓ for modelling the two data attributes independently
- $\frac{1}{2}$  ✓ for computing  $\pi$  correctly
- $\frac{1}{2}$  ✓ for computing  $\mu_1, \dots, \mu_3$  correctly
- $\frac{1}{2}$  ✓ for computing  $\alpha_1, \dots, \alpha_3$  correctly

b)

The unnormalized posterior is  $p(y^{(b)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_1^{(b)} | y^{(b)}) p(\mathbf{x}_2^{(b)} | y^{(b)}) p(y^{(b)})$ , so we evaluate that for all three choices of  $y^{(b)}$  and get

$$p(y^{(b)} | \mathbf{x}^{(b)}) \propto \left( e^{-\frac{9}{2}} \frac{1}{2} \frac{2}{7} \quad e^{-\frac{1}{2}} 0 \frac{2}{7} \quad e^{-\frac{9}{2}} \frac{2}{3} \frac{3}{7} \right)^T = \begin{pmatrix} \frac{1}{7e^{\frac{9}{2}}} & 0 & \frac{2}{7e^{\frac{9}{2}}} \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

c)

We do not know anything about this data point, so the posterior distribution is just the prior distribution.

$$p(y^{(c)} | \mathbf{x}^{(c)}) = p(y) = \begin{pmatrix} \frac{2}{7} & \frac{2}{7} & \frac{3}{7} \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

d)

Since we only know the feature  $x_2^{(d)}$ , we only condition on that and get  $p(y^{(d)} | \mathbf{x}^{(d)}) \propto p(\mathbf{x}_2^{(b)} | y^{(d)}) p(y^{(d)})$ .

$$p(y^{(d)} | \mathbf{x}^{(d)}) = \begin{pmatrix} \frac{1}{2} \frac{2}{7} & 1 \frac{2}{7} & \frac{1}{3} \frac{3}{7} \end{pmatrix}^T = \begin{pmatrix} \frac{1}{7} & \frac{2}{7} & \frac{1}{7} \end{pmatrix}^T$$

- ✓ for the correct answer with a brief justification. No ✓ without a justification. ✓ is also granted if the student found wrong parameters in a).

## Problem 4 (Version D) (6 credits)

|   |  |
|---|--|
| 0 |  |
| 1 |  |
| 2 |  |
| 3 |  |

a)

In a naive Bayes classifier, the features are independent, so we can choose a different probability distribution for each of them. We choose to model the continuous feature as a normal distribution,  $x_1 | y = c \sim \mathcal{N}(\mu_c, 1)$ . The discrete feature can be one of two values which we model with a Bernoulli distribution  $x_3 | y = c \sim \text{Bernoulli}(\alpha_c)$  where yes is 1, no is 0 and  $\alpha_c$  gives the success probability. The distribution of the classes  $y$  is a categorical distribution with parameter  $\pi$ ,  $y \sim \text{Categorical}(\pi)$ . The maximum likelihood estimates of the parameters are

$$\pi = \left( \frac{2}{7} \quad \frac{2}{7} \quad \frac{3}{7} \right)^T$$

$$\mu_1 = -2 \quad \mu_2 = 2 \quad \mu_3 = 4$$

$$\alpha_1 = \frac{1}{2} \quad \alpha_2 = 0 \quad \alpha_3 = \frac{2}{3}$$

- $\frac{1}{2} \checkmark$  for modelling the class prior
- $\checkmark$  for modelling the two data attributes independently
- $\frac{1}{2} \checkmark$  for computing  $\pi$  correctly
- $\frac{1}{2} \checkmark$  for computing  $\mu_1, \dots, \mu_3$  correctly
- $\frac{1}{2} \checkmark$  for computing  $\alpha_1, \dots, \alpha_3$  correctly

|   |  |
|---|--|
| 0 |  |
| 1 |  |

b)

The unnormalized posterior is  $p(y^{(b)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_1^{(b)} | y^{(b)}) p(\mathbf{x}_2^{(b)} | y^{(b)}) p(y^{(b)})$ , so we evaluate that for all three choices of  $y^{(b)}$  and get

$$p(y^{(b)} | \mathbf{x}^{(b)}) \propto \left( e^{-8\frac{1}{2}\frac{2}{7}} \quad e^{0\frac{2}{7}} \quad e^{-2\frac{2}{3}\frac{3}{7}} \right)^T = \left( \frac{1}{7e^8} \quad 0 \quad \frac{2}{7e^2} \right)^T$$

- $\checkmark$  for the correct answer with a brief justification. No  $\checkmark$  without a justification.  $\checkmark$  is also granted if the student found wrong parameters in a).

|   |  |
|---|--|
| 0 |  |
| 1 |  |

c)

We do not know anything about this data point, so the posterior distribution is just the prior distribution.

$$p(y^{(c)} | \mathbf{x}^{(c)}) = p(y) = \left( \frac{2}{7} \quad \frac{2}{7} \quad \frac{3}{7} \right)^T$$

- $\checkmark$  for the correct answer with a brief justification. No  $\checkmark$  without a justification.  $\checkmark$  is also granted if the student found wrong parameters in a).

|   |  |
|---|--|
| 0 |  |
| 1 |  |

d)

Since we only know the feature  $x_2^{(d)}$ , we only condition on that and get  $p(y^{(d)} | \mathbf{x}^{(b)}) \propto p(\mathbf{x}_2^{(b)} | y^{(d)}) p(y^{(d)})$ .

$$p(y^{(d)} | \mathbf{x}^{(d)}) = \left( \frac{1}{2}\frac{2}{7} \quad 1\frac{2}{7} \quad \frac{1}{3}\frac{3}{7} \right)^T = \left( \frac{1}{7} \quad \frac{2}{7} \quad \frac{1}{7} \right)^T$$

- $\checkmark$  for the correct answer with a brief justification. No  $\checkmark$  without a justification.  $\checkmark$  is also granted if the student found wrong parameters in a).

## Problem 5 (Version A) (2 credits)

|                          |   |
|--------------------------|---|
| <input type="checkbox"/> | 0 |
| <input type="checkbox"/> | 1 |
| <input type="checkbox"/> | 2 |

We will prove that  $f(\mathbf{x})$  is convex using convexity-preserving operations.

$\mathbf{a}^T \mathbf{x}$  is convex in  $\mathbf{x}$  and  $e^z$  is an increasing convex function. Therefore, their composition  $e^{\mathbf{a}^T \mathbf{x}}$  is convex in  $\mathbf{x}$ .

Similarly,  $-\mathbf{a}^T \mathbf{x}$  is convex in  $\mathbf{x}$ , so  $e^{-\mathbf{a}^T \mathbf{x}}$  is convex in  $\mathbf{x}$  as well.

$e^{\mathbf{a}^T \mathbf{x}} + e^{-\mathbf{a}^T \mathbf{x}}$  is a sum of convex functions, so it's also convex in  $\mathbf{x}$ .

Finally,  $\exp(e^{\mathbf{a}^T \mathbf{x}} + e^{-\mathbf{a}^T \mathbf{x}})$  is a composition of  $e^z$  (an increasing convex function) with another convex function.

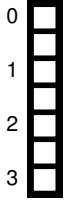
Therefore  $f(\mathbf{x})$  is convex in  $\mathbf{x}$ .

- 0.5 ✓ for Ansatz and some steps
- 0.5 ✓ for proving convexity of  $e^{\mathbf{a}^T \mathbf{x}}$
- 0.5 ✓ for applying sum rule
- 0.5 ✓ for correctly using the composition rule

0.5 ✓ are subtracted for each of the following mistakes:

- Not stating that  $\mathbf{a}^T \mathbf{x}$  and  $-\mathbf{a}^T \mathbf{x}$  are convex in  $\mathbf{x}$
- Not stating that  $e^z$  must be non-decreasing for the composition rule to apply
- Using the second derivative to prove that a function on  $\mathbb{R}^d$  is convex (instead of considering the Hessian)

## Problem 6 (Version A) (3 credits)



The output of conv1 will have shape [32, 16, 8]. Therefore,

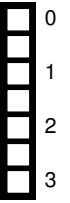
- $C_{in} = 32$  since the output of conv1 has 8 channels.
- $C_{out} = 16$  we know that the output of the NN has 16 channels.
- $P = 1$  and  $S = 1$  since no other combination of  $P$  and  $S$  will produce an output image with height 16 and width 8, since we don't need to perform downsampling in this layer.

0.5 ✓ for correct  $C_{in}$ , 0.5 ✓ for correct  $C_{out}$ , 0.5 ✓ for correct  $P$ , 1.5 ✓ for correct  $S$ . If no justification is provided (or if the justification is completely wrong), 0 points are given for the respective part.

Sample Solution

Correction Notes

## Problem 6 (Version B) (3 credits)



The output of conv1 will have shape [32, 64, 32]. Therefore,

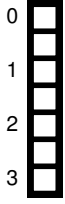
- $C_{in} = 32$  since the output of conv1 has 8 channels.
- $C_{out} = 16$  we know that the output of the NN has 16 channels.
- $P = 1$  (or  $P = 0$ ) and  $S = 4$  since no other combination of  $P$  and  $S$  will produce an output image with height 16 and width 8, i.e., where both dimensions are reduced by a factor of 4.

0.5 ✓ for correct  $C_{in}$ , 0.5 ✓ for correct  $C_{out}$ , 0.5 ✓ for correct  $P$ , 1.5 ✓ for correct  $S$ . If no justification is provided (or if the justification is completely wrong), 0 points are given for the respective part.

Sample Solution

Correction Notes

## Problem 6 (Version C) (3 credits)



The output of conv1 will have shape [8, 16, 8]. Therefore,

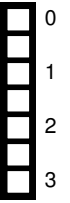
- $C_{in} = 8$  since the output of conv1 has 8 channels.
- $C_{out} = 16$  we know that the output of the NN has 16 channels.
- $P = 1$  and  $S = 1$  since no other combination of  $P$  and  $S$  will produce an output image with height 16 and width 8, since we don't need to perform downsampling in this layer.

0.5 ✓ for correct  $C_{in}$ , 0.5 ✓ for correct  $C_{out}$ , 0.5 ✓ for correct  $P$ , 1.5 ✓ for correct  $S$ . If no justification is provided (or if the justification is completely wrong), 0 points are given for the respective part.

Sample Solution

Correction Notes

## Problem 6 (Version D) (3 credits)



The output of conv1 will have shape [8, 64, 32]. Therefore,

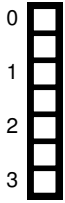
- $C_{in} = 8$  since the output of conv1 has 8 channels.
- $C_{out} = 16$  we know that the output of the NN has 16 channels.
- $P = 1$  (or  $P = 0$ ) and  $S = 4$  since no other combination of  $P$  and  $S$  will produce an output image with height 16 and width 8, i.e., where both dimensions are reduced by a factor of 4.

0.5 ✓ for correct  $C_{in}$ , 0.5 ✓ for correct  $C_{out}$ , 0.5 ✓ for correct  $P$ , 1.5 ✓ for correct  $S$ . If no justification is provided (or if the justification is completely wrong), 0 points are given for the respective part.

Sample Solution

Correction Notes

## Problem 7 (All Versions) (5 credits)



a)

Since  $\xi_q > 2$  the instance  $q$  is misclassified and lies on the wrong side of the decision boundary and it is *outside* of the margin. ✓

The vector  $\mathbf{w}_{\text{soft}}$  is a *feasible* solution for the new hard-margin SVM, i.e. it satisfies all of the constraints because:

- By removing instance  $q$  we remove the corresponding constraint
- All other instances  $i \neq q$  satisfy  $y_i(\mathbf{w}_{\text{soft}}^T \mathbf{x}_i + b) \geq 1$  since  $\xi_i = 0$

✓ Since we already found one feasible solution, namely  $\mathbf{w}_{\text{soft}}$  with the corresponding margin  $m_{\text{soft}} = \frac{2}{\|\mathbf{w}_{\text{soft}}\|}$ , the solution found by the hard-margin SVM with  $q$  removed can only be larger. ✓ Therefore,  $m_{\text{hard}} \geq m_{\text{soft}}$ . We generously allow answers  $m_{\text{hard}} \geq m_{\text{soft}}$  and  $m_{\text{hard}} = m_{\text{soft}}$ . The main objective was to realize that instance  $q$  is misclassified and outside of the margin.



b)

Since  $\xi_q > 2$  the instance  $q$  is misclassified and lies on the wrong side of the decision boundary and it is *outside* of the margin. ✓

As before, the vector  $\mathbf{w}_{\text{soft}}$  is a *feasible* solution for the new hard-margin SVM, i.e. it satisfies all of the constraints. The constraint for instance  $q$  before was  $y_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b) \geq 1 - \xi_q$ . The optimal solution for

$$\xi_q \begin{cases} 1 - y_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b), & \text{if } y_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b) < 1 \\ 0, & \text{otherwise} \end{cases}$$

$\xi_q > 2$  implies  $y_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b) < -1$ . If we now flip the sign of  $-y_q = \tilde{y}_q$ , we get  $\tilde{y}_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b) > 1$ . Hence,  $\tilde{\xi}_q = 0$  (instances  $q$  is now correctly classified and outside the margin). As before, all other instances  $i \neq q$  satisfy  $y_i(\mathbf{w}_{\text{soft}}^T \mathbf{x}_i + b) \geq 1$  since  $\xi_i = 0$ . ✓

Substituting  $\xi_q > 2$  we have  $y_q(\mathbf{w}_{\text{soft}}^T \mathbf{x}_q + b) \geq -1$ . By relabeling instance  $q$ , i.e. multiplying  $y_q$  by  $-1$  the hard-margin constraint is satisfied.

Since we already found one feasible solution, namely  $\mathbf{w}_{\text{soft}}$  with the corresponding margin  $m_{\text{soft}} = \frac{2}{\|\mathbf{w}_{\text{soft}}\|}$ , the solution found by the hard-margin SVM with  $q$  relabeled can only be larger or be as large. Therefore,  $m_{\text{hard}} \geq m_{\text{soft}}$  (we also accept  $m_{\text{hard}} = m_{\text{soft}}$ ).

Note that if a wrong answer for a) was given because of the assumption that point  $q$  is inside of the margin, this is considered to be an error propagation and allows for one point in b).

## Problem 8 (All Versions) (6 credits)

a)

The training error is 0. ✓ Since  $M'$  is a rank 1 matrix  $X'$  and  $y'$  are linearly dependent which means we can perfectly reconstruct  $y'$  from  $X'$ . ✓



b)

Since the training error is 0 as we reasoned above we have:  $w^* X' + b^* = y'$ . ✓

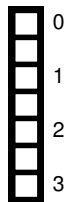
Since  $M'$  is the *best* rank 1 approximation of  $M$  we have:  $M' = \sigma_1 u_1 v_1^T$  where  $\sigma_1$  is the largest singular value, and  $u_1$  and  $v_1$  are the corresponding singular vectors.

From here we can conclude that  $X' = \sigma_1 u_1 v_{11}$  and  $y' = \sigma_1 u_1 v_{12}$  where  $v_{11}$  and  $v_{12}$  are the first and second element of  $v_1$  respectively. ✓ Plugging  $X'$  and  $y'$  in we have:

$$\begin{aligned} w^* X' + b^* &= y' \\ w^* \sigma_1 u_1 v_{11} + b^* &= \sigma_1 u_1 v_{12} \\ w^* v_{11} + b^* &= v_{12} \end{aligned}$$

From here we have:  $b^* = 0$  and  $w^* = \frac{v_{12}}{v_{11}}$ . ✓

This solution is not unique. So every solution that fulfills the equation above is considered to be correct.



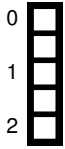
c)

Since we assume that  $X'$  is full rank there are only two valid options:  $K = D$  or  $K = D + 1$ . If  $K = D$  then  $y'$  can be expressed as a linear combination of  $X'$  and we again achieve an error of 0. If  $K = D + 1$  then the training error depends on the dataset and is in general  $\geq 0$ . ✓

Above, we made the simplifying assumption that  $D \geq N$ . However, the argument holds also for  $D < N$  by substituting  $D$  with  $N$ .



## Problem 9 (Version A) (6 credits)



a)

The objective for the (squared) Mahalanobis distance is  $J(\mathbf{X}, \mathbf{Y}, \boldsymbol{\mu}) = \sum_{i=1}^N \sum_{k=1}^K \mathbf{z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)$ . By considering the optimization  $\min_{\mathbf{z}} J(\mathbf{X}, \mathbf{Y}, \boldsymbol{\mu})$   $\frac{1}{2}$  ✓ we can directly see the cluster assignment update from this:

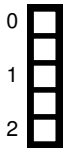
$$\mathbf{z}_{ik} = \begin{cases} 1 & \text{if } k = \arg \min_j (\mathbf{x}_i - \boldsymbol{\mu}_j)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_j) \\ 0 & \text{otherwise. } \frac{1}{2} \text{ ✓} \end{cases} \quad (1)$$

Using the objective we can also derive the centroid update as

$$\frac{\partial J}{\partial \boldsymbol{\mu}_k} = \frac{\partial}{\partial \boldsymbol{\mu}_k} \sum_{i=1}^N \mathbf{z}_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) = - \sum_{i=1}^N \mathbf{z}_{ik} 2 \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)^T = 0 \quad (2)$$

$$\Leftrightarrow \sum_{i=1}^N \mathbf{z}_{ik} \boldsymbol{\mu}_k = \sum_{i=1}^N \mathbf{z}_{ik} \mathbf{x}_i \Leftrightarrow \boldsymbol{\mu}_k = \frac{\sum_{i=1}^N \mathbf{z}_{ik} \mathbf{x}_i}{\sum_{i=1}^N \mathbf{z}_{ik}} \text{ ✓} \quad (3)$$

Interestingly, the Mahalanobis distance does not have an influence on the centroid update.



b)

Version A. This solution is much more thorough than necessary. A good reason for the diagonal terms or the identity matrix is awarded up to 1.5 points, depending on its precision and completeness (0 points if no reason is given). The last 0.5 points are awarded if the reasoning covers the off-diagonal terms.

Denote  $\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}$ . The boundary between  $\boldsymbol{\mu}_1$  and  $\boldsymbol{\mu}_2$  is  $\mathbf{x} = (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2)/2 + c(0, 1)^T$  for  $c \geq 0$ . For any boundary we have  $d(\mathbf{x}, \boldsymbol{\mu}_1) = d(\mathbf{x}, \boldsymbol{\mu}_2)$ . We thus have

$$(\mathbf{x} - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_1) = (\mathbf{x} - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_2) \Leftrightarrow \begin{pmatrix} 1 & c \end{pmatrix} \boldsymbol{\Sigma}^{-1} \begin{pmatrix} 1 \\ c \end{pmatrix} = \begin{pmatrix} -1 & c \end{pmatrix} \boldsymbol{\Sigma}^{-1} \begin{pmatrix} -1 \\ c \end{pmatrix} \quad (4)$$

$$\Leftrightarrow \sigma_{11} + 2c\sigma_{12} + c^2\sigma_{22} = \sigma_{11} - 2c\sigma_{12} + c^2\sigma_{22}$$

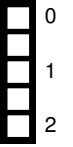
and therefore  $\sigma_{12} = 0$ . The boundary between  $\boldsymbol{\mu}_1$  and  $\boldsymbol{\mu}_3$  is  $\mathbf{x} = (\boldsymbol{\mu}_1 + \boldsymbol{\mu}_3)/2 + c(1, 1)^T$  for a certain range of  $c$ . Considering  $\sigma_{12} = 0$  and  $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_3 = (1, -1)^T$  we have

$$(c + 0.5)^2 \sigma_{11} + (c - 0.5)^2 \sigma_{22} = (c - 0.5)^2 \sigma_{11} + (c + 0.5)^2 \sigma_{22} \quad (5)$$

and thus  $\sigma_{11} = \sigma_{22}$ . Since  $\boldsymbol{\Sigma}$  is PSD and invertible,  $\boldsymbol{\Sigma}^{-1}$  must be PD. We therefore have (for any  $a > 0$ )

$$\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}. \quad (6)$$

c)



Version A. This solution is much more thorough than necessary. A good reason for the diagonal terms is awarded up to 1.5 points, depending on its precision and completeness (0 points if no reason is given). The last 0.5 points are awarded if the reasoning covers the off-diagonal terms.

Since there is a vertical/horizontal boundary in the center we have  $\sigma_{12} = 0$  (see previous subproblem). The boundary between  $\mu_2$  and  $\mu_3$  is  $\mathbf{x} = (\mu_2 + \mu_3)/2 + c(2, 1)^T$  for a certain range of  $c$ . With  $\mu_2 - \mu_3 = (1, -1)^T$  we therefore have

$$\begin{aligned} (2c + 0.5 \quad c - 0.5) \Sigma^{-1} \begin{pmatrix} 2c + 0.5 \\ c - 0.5 \end{pmatrix} &= (2c - 0.5 \quad c + 0.5) \Sigma^{-1} \begin{pmatrix} 2c - 0.5 \\ c + 0.5 \end{pmatrix} \\ \Leftrightarrow (4c^2 + 2c + 0.25)\sigma_{11} + (c^2 - c + 0.25)\sigma_{22} &= (4c^2 - 2c + 0.25)\sigma_{11} + (c^2 + c + 0.25)\sigma_{22}. \end{aligned} \quad (7)$$

Considering only terms with  $c^1$  we have  $2\sigma_{11} - \sigma_{22} = -2\sigma_{11} + \sigma_{22} \Leftrightarrow 4\sigma_{11} = 2\sigma_{22}$ .

Since a covariance matrix is PSD and  $\Sigma$  is invertible,  $\Sigma^{-1}$  must be positive definite. the solution for each version is (for any  $a > 0$ )

$$\Sigma_A^{-1} = \begin{pmatrix} a & 0 \\ 0 & 2a \end{pmatrix}, \quad \Sigma_B^{-1} = \begin{pmatrix} 2a & 0 \\ 0 & a \end{pmatrix}, \quad \Sigma_C^{-1} = \begin{pmatrix} a & 0 \\ 0 & 2a \end{pmatrix}, \quad \Sigma_D^{-1} = \begin{pmatrix} 2a & 0 \\ 0 & a \end{pmatrix}. \quad (8)$$

## Problem 10 (Version A) (6 credits)

|   |  |
|---|--|
| 0 |  |
| 1 |  |
| 2 |  |
| 3 |  |

a)

Changing any single instance only modifies one of the groups  $G_i$  so it is sufficient to reason *only* about the sensitivity of the aggregation function operating on the groups. ✓

Since  $f$  is bounded, the aggregation function takes as input  $m$  numbers,  $g_1, \dots, g_m$  in the interval  $[a, b]$ . Changing one instance can change at most one  $g_i$ , and in the worst case the change can be anywhere in the interval  $[a, b]$ . ✓

In the worst-case the output of one  $g_i$  changes from  $b$  to  $a$ , and the global  $\Delta_1$  sensitivity of  $f'$  is  $\frac{b-a}{m}$ . ✓

- 1 ✓ for stating that changing one instances changes only one group and thus one  $g_i$ .
- 1 ✓ for stating that we should reason about the aggregation function for  $m$  inputs bounded in  $[a, b]$ .
- 1 ✓ for correctly computing the worst-case sensitivity.

|   |  |
|---|--|
| 0 |  |
| 1 |  |
| 2 |  |

b)

The global sensitivity of  $f'$  does not depend on  $n$  and therefore does not change. ✓

The global sensitivity of  $f'$  decreases as we increase  $m$  since we are dividing by  $m$ . ✓

- 1 ✓ for stating that it does not depend on  $n$ .
- 1 ✓ for stating the correct dependence on  $m$ .
- Full points for correct answers regardless of whether a) was solved.

|   |  |
|---|--|
| 0 |  |
| 1 |  |

c)

We can obtain  $\epsilon$ -DP by adding noise from the Laplace distribution with zero mean and variance  $\frac{\Delta_1}{\epsilon}$  where  $\Delta_1 = \frac{b-a}{m}$ . ✓

- 1 ✓ for the correct noise distribution with correct mean and variance.
- 0.5 ✓ if the mean or the variance is not specified or if they are wrong.
- 0.5 ✓ if the wrong distribution is specified.
- 0.5 ✓ if the correct sensitivity is not provided either here or in a).

## Problem 10 (Version B) (6 credits)

a)

Changing any single instance only modifies one of the groups  $G_i$  so it is sufficient to reason *only* about the sensitivity of the aggregation function operating on the groups. ✓

Since  $f$  is bounded, the aggregation function takes as input  $m$  numbers,  $g_1, \dots, g_m$  in the interval  $[a, b]$ . Changing one instance can change at most one  $g_i$ , and in the worst case the change can be anywhere in the interval  $[a, b]$ . ✓

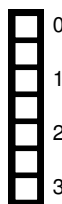
In the worst-case we have the following scenario:

Before changing a single instance:  $g_1 = a, g_2 = a, \dots, g_{m/2} = a, g_{m/2+1} = b, \dots, g_{m-1} = b, g_m = b$

After changing a single instance:  $g_1 = a, g_2 = a, \dots, g_{m/2} = b, g_{m/2+1} = b, \dots, g_{m-1} = b, g_m = b$

Here the median is  $g_{m/2}$  and it has changed from  $a$  to  $b$ . Therefore, the global  $\Delta_1$  sensitivity of  $f'$  is  $b - a$ . ✓

- 1 ✓ for stating that changing one instances changes only one group and thus one  $g_i$ .
- 1 ✓ for stating that we should reason about the aggregation function for  $m$  inputs bounded in  $[a, b]$ .
- 1 ✓ for correctly computing the worst-case sensitivity.



b)

The global sensitivity of  $f'$  does not depend on  $n$  and therefore does not change. ✓

The global sensitivity of  $f'$  does not depend on  $m$  and therefore does not change. ✓

- 1 ✓ for stating that it does not depend on  $n$ .
- 1 ✓ for stating the correct dependence on  $m$ .
- Full points for correct answers regardless of whether a) was solved.



c)

We can obtain  $\epsilon$ -DP by adding noise from the Laplace distribution with zero mean and variance  $\frac{\Delta_1}{\epsilon}$  where  $\Delta_1 = b - a$ . ✓

- 1 ✓ for the correct noise distribution with correct mean and variance.
- 0.5 ✓ if the mean or the variance is not specified or if they are wrong.
- 0.5 ✓ if the wrong distribution is specified.
- 0.5 ✓ if the correct sensitivity is not provided either here or in a).



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

The image shows a large grid of graph paper, intended for writing solutions. A diagonal watermark is overlaid across the grid. The watermark consists of two parts: 'Sample Solution' in a large, light blue font, and 'Correction Notes' in a smaller, red font below it. The grid is composed of small squares, typical of graph paper used in mathematics or engineering.

Sample Solution  
Correction Notes

Sample Solution  
Correction Notes